

## **gamma-Irradiation dose: effects on baby-leaf spinach ascorbic acid, carotenoids, folate, alpha-tocopherol, and phyloquinone concentrations.**



Ionizing radiation of fruits and vegetables, in the form of gamma rays or electron beams, is effective in overcoming quarantine barriers in trade and prolonging shelf life, but a void of information persists on ionizing radiation effects of vitamin profiles in individual foods. Baby-leaf spinach from commercial cultivars, flat-leafed 'Lazio' and crinkled-leaf 'Samish', was grown, harvested, and surface sanitized according to industry practices. Baby-leaf spinach of each cultivar was packaged under air or N(2) atmosphere, representing industry practices, then exposed to cesium-137 gamma-radiation at 0.0, 0.5, 1.0, 1.5, or 2.0 kGy. Following irradiation, leaf tissues were assayed for vitamin (C, E, K, B(9)) and carotenoid (lutein/zeaxanthin, neoxanthin, violoxanthin, and beta-carotene) concentrations. Atmospheres by irradiation had little consistent effect, but N(2) versus air was associated with elevated dihydroascorbic acid levels. Four phytonutrients (vitamins B(9), E, and K and neoxanthin) exhibited little or no change in concentration with increasing doses of irradiation. However, total ascorbic acid (vitamin C), free ascorbic acid, lutein/zeaxanthin, violaxanthin, and beta-carotene all were significantly reduced at 2.0 kGy and, depending on cultivar, were affected at lesser doses of 0.5 and 1.5 kGy. Dihydroascorbic acid, the most affected compound and an indicator of stress, likely due to irradiation-generated oxidative radicals, increased with increasing irradiation doses >0.5 kGy.